

Notes Service

Developer 2 | Structure & dependencies | M1/M2 contract

Purpose and ownership

Owns curated categories, notes, draft ownership, editorial publishing and immutable revisions in MongoDB. Community blogs and subscriptions belong to other services.

Dependencies

Java 21; Maven Wrapper; pinned Spring Boot 3.5.x + compatible Spring Cloud; Web MVC; Spring Data MongoDB; OAuth2 Resource Server; Validation; Actuator; JUnit + Testcontainers. M2 publication needs Mongo replica set.

Proposed package / class map

PACKAGE	CLASSES AND RESPONSIBILITY
config / security	MongoConfig, SecurityConfig, JwtClaimsConverter - connectivity, JWT issuer/audience, roles/scopes
category/controller + dto	CategoryController; CreateCategoryRequest, PatchCategoryRequest, CategoryResponse
category/service + repository	CategoryService, CategoryRepository, CategoryDocument - recursive tree, indexes, ancestor rules
note/controller + dto	NoteController; CreateNoteRequest, PatchNoteRequest, NoteResponse, NoteListItemResponse
note/service + repository	NoteService, NoteRepository, NoteDocument - owner-filtered draft CRUD and ETags
publication / revision (M2)	PublicationService, RevisionRepository, NoteRevisionDocument - review, atomic snapshot, public reads
common/error	ApiExceptionHandler, ApiErrorResponse, PageResponse - standard responses

Service links

React -> Gateway :8080 -> Notes :8081 -> MongoDB; Notes validates User/OAuth :9000 JWKS. One recursive categories collection; primaryCategoryId; JWT sub = authorId. Public API reads snapshot only.

Build order

B0 baseline -> M1 category tree + secure draft CRUD -> M2 submit/reject/publish/revisions/public reads -> M2.5 tiered access and Billing entitlement -> M3 interview Q&A.

Implementation note

M1 accepts NOTE/DRAFT only. Existing PRIVATE/PUBLIC visibility is not yet the three-tier access model. Older separate topics schema is superseded pending review.